

Simone Carletti

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About Me

I am a **Computer Engineering M.Sc.** student specializing in **robotics and autonomous systems**. I'm currently completing my master's thesis at EPFL on adaptive control for a **morphing-wing drone**, with a focus on **agile indoor flight**. I have hands-on experience in **perception, planning, control systems, ROS2, and embedded programming**. I am proficient in **C++**, **Python**, and Typescript, with research experience at EPFL and DTU. Outside the lab, I enjoy dancing salsa, hiking, taking pictures, and exploring new music.

Education

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| EPFL - LIS, Master's Thesis <ul style="list-style-type: none">Title: L1-Adaptive augmentation for robust morphing-wing drone flight controlSkills: Control Systems, C++, Python, Simulink, ROS2, Git, Linux, Unity | <i>Sept 2025 – Present</i> |
| DTU, M.Sc. Autonomous Systems <ul style="list-style-type: none">Erasmus+ exchange | <i>Sept 2024 – Jul 2025</i> |
| Politecnico di Torino, M.Sc. Computer Engineering <ul style="list-style-type: none">Curriculum in Automation and Intelligent Cyber-Physical Systems - Current GPA: 29.48/30 | <i>Sept 2023 – Present</i> |
| Università degli Studi di Firenze, B.Sc. Computer Science and Engineering <ul style="list-style-type: none">Graduated with 110/110 cum laude (with honors)Thesis: Sensor network to monitor the greenhouses of the Botanical Garden of Florence | <i>Sept 2020 – Sept 2023</i> |

Experience

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| Robotic Engineer, DIANA student team <ul style="list-style-type: none">Contributed to a student project developing a Mars-inspired rover, learning and applying robotics fundamentals within a team environmentExplored perception and EKF-based state estimation concepts using C++Participated in software development tasks, testing sessions, and system integration activities | <i>Sept 2023 – Jul 2024
Turin, Italy</i> |
| Full-Stack Developer, Genomedics S.r.l <ul style="list-style-type: none">Lead developer of three key projects that handle the release, deployment, and licence management for all the company's products. Developer for the company's main project (KPlatform).Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux | <i>Sept 2020 – Present
Florence, Italy</i> |

Selected Projects

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|---|-----------------------------|
| Digital Control - Autonomous Robotics <ul style="list-style-type: none">Designed digital controllers and implemented them on embedded microcontrollers (Github 🔗) | <i>Sept 2024 – Dec 2024</i> |
| Autonomous Navigation & Task Execution - DTU RoboCup <ul style="list-style-type: none">Developed autonomous navigation, control, and task execution algorithms with the Robobot platform for the DTU RoboCup competition | <i>Feb 2025 – May 2025</i> |
| Localization with EKF in ROS2 <ul style="list-style-type: none">Implemented a full EKF localization pipeline from scratch in ROS2 using Python | <i>Nov 2023 – Dec 2023</i> |

Misc.

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| TuttoMeritoMio Excellence Scholarship <ul style="list-style-type: none">Scholarship dedicated to the top 100 students from the University of Florence | <i>2020 - 2023</i> |
| ESN (Copenhagen and DTU) Volunteer <ul style="list-style-type: none">Organized events, supported international students and promoted cultural exchange. | <i>Jan 2025 - Jul 2025</i> |

Skills

- Languages:** Fluent in English, native Italian speaker, basic Spanish skills
- Programming:** TypeScript, JavaScript, C, C++, Python, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, soldering, PyTorch, Google Colab, OpenCV, Nvidia Jetson/Orin, Onshape
- Certificates:** Cambridge B2 (FCE), ID: C4570372. ECDL Full Standard. ID: IT 2156394